Lecture 2

From Sample to Sequence

Day 1 · 09:45–10:30 · 45 min

LO2: describe Illumina and Nanopore outputs and choose between them · LO6 (part): understand the outsourcing workflow

The journey of an isolate

Wound swab → **Culture** → **Pure isolate**

DNA extraction → **Library prep** → **Sequencing** → **Data files** → **Analysis**

KEY MESSAGE

Everything up to the pure isolate (dark boxes) already happens in our labs.

We sequence isolates, not swabs

- An **isolate** is a pure culture of one bacterium, so its genome is clean
- Sequencing a swab directly (**metagenomics**) mixes human DNA with many microbes: harder and more expensive
- For SSI surveillance, **isolate sequencing is the standard**

> **KEY MESSAGE**
>
> **Store your isolates (–80 °C or glycerol stocks). You can't sequence what you threw away.**

DNA extraction and library preparation

- **Extraction:** break open the cells and purify the DNA
- **Library prep:** cut or tag the DNA and add short “adapter” sequences so the machine can read it
- **Multiplexing:** many samples are barcoded and run together, which lowers the cost per genome

> You don’t need to do this yourself. But poor-quality DNA gives poor-quality data (Lab 2).

Two main technologies

	Illumina (short reads)	Oxford Nanopore (long reads)
Read length	100–300 letters, in pairs	thousands to 100,000+ letters
Accuracy	very high	now high with modern chemistry
Best for	SNP-level outbreak work	complete genomes, **plasmids**
Machine	large, central labs	MinION is pocket-sized
Files	FASTQ pair (R1 + R2)	one FASTQ

Jigsaw analogy: Illumina gives many tiny pieces; Nanopore gives fewer, bigger pieces.

What is a read?

- A **read** is one piece of DNA sequence produced by the machine
- One bacterial genome gives **hundreds of thousands to millions** of reads
- We put the reads back together into the genome

> Like rebuilding a shredded book from many overlapping copies.

Coverage (depth)

Coverage = how many reads, on average, cover each position

$$(\text{number of reads} \times \text{read length}) \div \text{genome size}$$

$$1{,}000{,}000 \times 150 \div 5{,}000{,}000 = \mathbf{30\times}$$

KEY MESSAGE

Aim for ≥ 30× with Illumina and ≥ 40–50× with Nanopore.

Which should I use?

Question	Best choice
Is this an outbreak? (fine SNP differences)	Illumina
Is resistance on a plasmid moving between species?	Nanopore or hybrid
Need results fast, locally	Nanopore
Lowest cost per genome at large scale	Illumina (batched)
Best of both	**Hybrid** assembly

KEY MESSAGE

Your sequencing partner may use either. This course teaches both.

Outsourcing: how it works

Agree on a partner → **Ship isolates or DNA** → **Partner sequences** → **You analyse**

- **Partners:** national reference lab, Africa CDC network, academic collaborator, or a commercial service
- **Shipping:** follow their biosafety rules; you may need permits and a material transfer agreement
- **You get back:** FASTQ files, sometimes assemblies and reports. **Keep the data.**

Questions to ask a sequencing partner

Platform?
Illumina, Nanopore, or both

Coverage?
≥ 30× Illumina

Raw reads?
Make sure FASTQ files are included

Turnaround?
Weeks from shipping to data

Cost?
Per isolate, including shipping

Data sharing?
Who owns and publishes the data

Metadata: the most underrated part

A genome without **when, where and from whom** is almost useless for surveillance.

Minimum fields

- isolate ID · collection date
- facility · ward · specimen type
- surgery type and date
- AST results

PROTECT PRIVACY

Never put names or patient IDs in shared files. Use coded study IDs.

Standard: **PHA4GE** contextual data specification

What files come back?

File	What it is
`sample_R1.fastq.gz` + `sample_R2.fastq.gz`	Illumina reads (a pair)
`sample.fastq.gz`	Nanopore reads
`sample.fasta`	Assembly (sometimes)
QC report	Sometimes

> In Lab 1 we'll open these files and look inside.

Quick check (show of hands)

1. You want to know if **6 MRSA isolates from one ward** are an outbreak. Which data would you request?

2. You suspect a **carbapenemase gene is jumping** between *Klebsiella* and *E. coli*. Which data would you request?

Quick check: answers

1. **Illumina, ≥ 30×**, with collection dates and ward metadata

2. **Nanopore or hybrid**, to resolve the plasmids